Ananya Balagere

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EDUCATION

University of California San Diego, San Diego, CA. — ***Expected Graduation 2028***
Master of Science in Computer Science

University of California Santa Cruz, Santa Cruz, CA. GPA: 3.88, Cum Laude — **June 2026**
Bachelor of Science in Computer Science
Relevant courses: Data Structures and Algorithms, Computer Networks, Distributed Systems, Socket Programming, Computer Architecture, Algorithm Analysis, Compilers

Bachelor of Science in Mathematical Theory and Computation
Relevant courses: Real Analysis, Partial Differential Equations, Linear Algebra, Cryptography, Chaos Theory
Undergraduate Thesis Topic: Schrodinger's Equation as an Introduction to Quantum Mechanics

EXPERIENCE

Software Development Intern @ Amazon (Language: Java) — **June 2026 - September 2026**
- Building an AI powered code review analyzer using Amazon Bedrock AgentCore, AWS S3, and Lambda infrastructure that checks code changes against product requirements and generates HTML reports for each review.
- Developing and pitching a prototype for an issue reporting tool to support account managers servicing Amazon Ads customers.
- Utilizing Chrome extension software and Amazon Bedrock AgentCore to build the tool, enabling users to capture screenshots and chat with an agent to describe their issue.

Software Development Intern @ Amazon (Languages: Javascript, Typescript, JSONata) — **September 2025 - December 2025**
- Implemented a query scheduling system for Amazon Publisher Cloud, a service provided for third-party advertisement publishers using AWS Lambda, AWS Clean Rooms, AWS Step Functions, and Amazon DynamoDB.
- Reduced overall latency and optimized for the amount of resources utilized during query execution.

Software Engineering Intern @ Apple (Language: Python) — **June 2025 - September 2025**
- Implemented a feature for Apple's internal testing platform for operating systems that handles device panics during testing on an Apple device and allows testing to continue.
- Cross-functional feature that cut weeks from the overall pipeline due to unnecessary testing.
- Improved the error classification and reporting for users of the testing platform.

Undergraduate Research Assistant (Language: Rust) — **June 2024 - February 2025**
Center for Research in Systems and Storage
- Worked with graduate students and industry specialists to develop a memory stick with an emphasis on secure deletion.
- Contributed to the open source Twizzler Operating System.
- Wrote a shim layer for a thread-safe implementation of TcpListener and TcpStream.
- DARPA Project W912CG-22-C-0011 TWISTED
- Presented a poster at UCSC's Association for Computing Machinery Undergrad Conference

Command and Data Handling Engineer — **September 2024 - December 2024**
NASA L'SPACE Mission Concept Academy (Workforce Preparation)
- Contributed to the writing of NASA mission life cycle documents up to the Preliminary Design Review. Contributed to the design of a robotics mission.
- Led the design of the Command and Data Handling Subsystem of a spacecraft to investigate lunar pit caves for sustainability of life.

- Performed trade studies between subsystem architectures to determine which instrument would best satisfy the mission's science objectives. **September 2023 -**
- Designed the software workflow for the spacecraft. **June 2025**

Learning Support Services Large Group Tutor (Languages: C, Haskell)
University of California Santa Cruz, Santa Cruz, CA

- Tutored 20 of my peers 3 times a week in various classes, including Introduction to Computer Systems and C Programming, Discrete Mathematics, and Introduction to Programming Languages.
- Helped with programming assignments, explaining concepts, and preparing for exams.

PROJECTS

Causally Consistent, Eventually Consistent, and Available Distributed Key-Value Store (Rust)

- Shards data into partitions and minimizes its movement through consistent hashing. Eventually consistent after 10 seconds. Utilizes Rust's tokio asynchronous runtime.

Mood Tracker/Recommender (React.js, Firebase, HTML, CSS) Web application to track and improve moods

- Includes a calendar to track moods, offers mood improvement recommendations.

Secure HTTP Proxy Server (C) Robust, Thread-Safe and Multithreaded

- Securely proxies HTTP requests to servers not forbidden by the user. Utilizes the OpenSSL library.

Reliable UDP Client and Server (C)

- Reliable file transfer between client and server implemented over UDP.

Course Planner (React.js, CSS) Web application that can generate potential four-year course plans for UCSC students

- Used Google Firebase, ReactJS, CSS, and Beautiful Soup. Created for the Google Solution Challenge.

Web Crawler (C) Crawls through the internet given a starting URL

- Implementation of linked lists, hashtable, and various C libraries. Using files and directories to save crawled pages.

SKILLS

- C, C++, Python, Java, Rust, Javascript, Unix/Linux, Git
- Amazon Web Services (AWS Lambda, AWS Step Functions, AWS Clean Rooms, S3), Amazon DynamoDB, Amazon Bedrock AgentCore
- Scrum, Jira, Canva, MATLAB, Figma, Web Development Languages (HTML, CSS, React.js, Node.js)
- Components of Adobe (Acrobat, Illustrator, Photoshop), Microsoft Suite (Teams, Excel, Word, OneNote)

MEMBERSHIPS

- UCSC Robotics, UCSC Google Developer Student Clubs, UCSC Association for Computing Machinery, UCSC Chamber Singers

HONORS

- Dean's Honors awarded to students in the top 15% of their class
  - Received Fall 2022, Winter 2023, Spring 2023, Fall 2023, Winter 2024, Fall 2024, Winter 2025, Winter 2026, Spring 2026